Practice Assignment 5 - Not Graded

This assignment will not be graded and is only for practice.
Multiple Choice Questions (MCQ):

1 point
Let AA and BB be square matrices of order 2. Which of the following option is true?
○
$nullity(AB) = nullity(A) + nullity(B)$nullity(AB) = nullity(A) + nullity(B)
○
$nullity(AB) = nullity(A)nullity(B)$nullity(AB) = nullity(A)nullity(B)
○
$nullity(A - B) = nullity(A) - nullity(B)$nullity(A − B) = nullity(A) − nullity(B)
○
$Rank(A + B) = Rank(A) + Rank(B)$Rank(A + B) = Rank(A) + Rank(B)
○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
None of the above.

Multiple Select Questions (MSQ):

1 point
A molecule is composed of atoms. A molecule of Ethylene, with the chemical formula C_2H_4C2H4, consists of two Carbon atoms and four Hydrogen atoms. A molecule of Oxygen, with the formula O_2O2 , consists of two Oxygen atoms. Note that Carbon, Hydrogen and Oxygen are denoted by the letters $C, H,$ and OC, H, and O respectively in the formula. When Ethylene comes in contact with Oxygen (O_2O2); Carbon dioxide (CO_2CO2) and water (H_2OH2O) are produced as the products of the chemical reaction . The equation corresponding to the chemical reaction (R) is given below

$$x_1C_2H_4 + x_2O_2 \longrightarrow x_3CO_2 + x_4H_2O.$$ … x1C2H4 + x2O2 ⟶ x3CO2 + x4H2O. … (R)

To balance the chemical equation we have to choose $x_1, x_2, x_3,$ and x_4 x1, x2, x3, and x4 such that both sides have the same number of carbon atoms on each side, the same number of hydrogen atoms on each side, and the same number of oxygen atoms on each side.
Note: An example to write the system of linear equations for balancing the chemical equation is the following :

$x_1C + x_2O \longrightarrow x_3CO_2$ x1C + x2O ⟶ x3CO2
corresponding system of linear equations is: $x_1 = x_3$ $x_2 = 2x_3$
corresponding system of linear equations is: x1 = x3x2 = 2x3

Consider the system of linear equations obtained for balancing the chemical equation (R) to answer the question.
Which of the following statements are true?
☐ The nullity of the matrix corresponding to this system is 0.
☐ The nullity of the matrix corresponding to this system is 1.
☐

$\{(1,3,1,1)\}$ is a basis of the null space of the matrix corresponding to this system.

- [ ] There are an infinite number of ways to balance the chemical equation (R).
- [ ] There is a unique way to balance the chemical equation (R).

No, the answer is incorrect.

Score: 0

Accepted Answers:

The nullity of the matrix corresponding to this system is 1.

There are an infinite number of ways to balance the chemical equation (R).

1 point

Which of the following option(s) is(are) true?

- [ ] If $A_{2\times 3}$ is a non zero matrix, then nullity of the matrix ≤ 2
- [ ] If $A_{3\times 2}$ is a non zero matrix, then nullity of the matrix ≤ 1.
- [ ] Let A and B be two square matrices of order 3, If nullity of matrix AB is 0, then nullity of matrix A is also zero.
- [ ] Let A and B be two square matrices of order 3, If nullity of matrix AB is 0, then nullity of matrix B is also zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $A_{2\times 3}$ is a non zero matrix, then nullity of the matrix ≤ 2

If $A_{3\times 2}$ is a non zero matrix, then nullity of the matrix ≤ 1.

Let A and B be two square matrices of order 3, If nullity of matrix AB is 0, then nullity of matrix A is also zero.

Let A and B be two square matrices of order 3, If nullity of matrix AB is 0, then nullity of matrix B is also zero.

1 point

Let $T: R^3 \rightarrow R^3$ be a linear transformation such that

$T(1,0,0) = (1,1,1), T(1,1,0) = (0,0,1)$, $T(1,0,0) = (1,1,1), T(1,1,0) = (0,0,1)$ and

$T(1,1,1) = (0,1,1)$, $T(1,1,1) = (0,1,1)$. Then which of the following option(s) is/ are true?

- [ ] $T(x,y,z) = (x-y, x-y+z, y)$
- [ ] $T(x,y,z) = (x-y, x-y+z, x)$
- [ ] $T(x,y,z) = (x+y, x-y+z, x)$
- [ ] T is one-one.
- [ ] T is not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x,y,z) = (x-y, x-y+z, x)$

T is one-one.

1 point

Consider a system of linear equations defined as follows:

$$\begin{aligned} x + y - z + w &= 0 \\ 2x + y + 3z - w &= 0 \\ 3x + 2y + 2z &= 0 \end{aligned}$$ x + y − z + w2x + y + 3z − w3x + 2y + 2z = 0 = 0 = 0

If AA is the coefficient matrix of the system of linear equations, then which of the following options is/are true?

- [ ] Null space of AA = $span\{(-4, 5, 1, 0), (2, -3, 0, 1)\}$span{(−4, 5, 1, 0), (2, −3, 0, 1)}
- [ ] Null space of AA = $span\{(-6, 8, 1, -1), (-4, 5, 1, 0)\}$span{(−6, 8, 1, −1), (−4, 5, 1, 0)}
- [ ] Null space of AA = $span\{(-3, 5, 1, 0), (2, -3, 0, 1)\}$span{(−3, 5, 1, 0), (2, −3, 0, 1)}
- [ ] Null space of AA = $span\{(-4, 5, 1, -1), (6, 7, 2, 1)\}$span{(−4, 5, 1, −1), (6, 7, 2, 1)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
Null space of AA = $span\{(-4, 5, 1, 0), (2, -3, 0, 1)\}$span{(−4, 5, 1, 0), (2, −3, 0, 1)}
Null space of AA = $span\{(-6, 8, 1, -1), (-4, 5, 1, 0)\}$span{(−6, 8, 1, −1), (−4, 5, 1, 0)}

Numerical Answer Type (NAT):
If nullity of a matrix $A_{5\times 7}$A5×7 is 2, then find the rank of the matrix $A_{5\times 7}$A5×7.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point

Let AA be a non-zero matrix of order $3 \times 13 \times 1$. What is the rank of the matrix AA^T AAT ?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Comprehension Type Question:

Ayesha, Pritha, Sabya, and Wang went on a trip to Manali and Kasol. Accommodation costs ₹1500 per day in Manali and ₹800 per day in Kasol. The total food cost is ₹2000 per day in Manali and ₹1200 per day in Kasol. They plan to spend 2 days in Manali and 2 days in Kasol. The first and second rows of the Table M2W5P1 shows the percentage of contribution by each of them for the accomodation at Manali and Kasol, respectively. Similarly, the first and the second row of the Table M2W5P2 shows the percentage of contribution by each of them for the food at Manali and Kasol, respectively.

Suppose $T(x, y)$T(x, y) denotes the contribution of a person for accommodation per day, where the first

variable xx denotes the percentage of contribution by that person for accommodation in Manali and the second variable yy denotes the percentage of contribution by that person for accommodation in Kasol. (i.e., if aa and bb denote the costs for accommodation per day at Manali and Kasol, respectively, then $T(x, y) = \frac{1}{100}(ax + by)$T(x, y) = 1001(ax + by)). Similarly, $T^{'}(v, w)$T′(v, w) denotes the contribution by a person for food per day, where the first variable vv denotes the percentage of contribution for food in Manali and the second variable ww denotes the percentage of contribution for food in Kasol. Answer the following questions based on the given information.

1 point
Choose the set of correct options.

- [ ] $T(x, y) = 30x + 16y$T(x, y) = 30x + 16y
- [ ] $T(x, y) = 15x + 8y$T(x, y) = 15x + 8y
- [ ] $T^{'}(v, w) = 20v + 12w$T′(v, w) = 20v + 12w
- [ ] $T^{'}(v, w) = 40v + 24w$T′(v, w) = 40v + 24w

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y) = 15x + 8y$T(x, y) = 15x + 8y
$T^{'}(v, w) = 20v + 12w$T′(v, w) = 20v + 12w

1 point
Choose the set of correct options.

- [ ] Suppose Ayesha contributes for herslef and also on behalf of Sabya. Then the contribution per day by Ayesha is given by $T(x_1 + x_3, y_1 + y_3)$T(x1 + x3, y1 + y3) and $T^{'}(v_1 + v_3, w_1 + w_3)$T′(v1 + v3, w1 + w3), for accommodation and food, respectively.
- [ ] The total contribution (for the whole trip) for the accommodation by Pritha is given by $2T(x_2, y_2)$ 2T(x2, y2), which is equal to $T(2x_2, 2y_2)$T(2x2, 2y2).
- [ ] The total contribution (for the whole trip) for the accommodation by Pritha is given by $2T(x_2, y_2)$ 2T(x2, y2), which is not equal to $T(2x_2, 2y_2)$T(2x2, 2y2).
- [ ] Suppose Pritha contributes for herself and also on behalf of Wang for food. Then the contribution per day by Pritha for food is given by $T^{'}(v_2, w_2) + T^{'}(v_4, w_4)$T′(v2, w2) + T′(v4, w4), which is not equal to $T^{'}(v_2 + v_4, w_2 + w_4)$T′(v2 + v4, w2 + w4).
- [ ] Suppose Pritha contributes for herself and also on behalf of Wang for food. Then the contribution per day by Pritha for food is given by $T^{'}(v_2, w_2) + T^{'}(v_4, w_4)$T′(v2, w2) + T′(v4, w4), which is equal to $T^{'}(v_2 + v_4, w_2 + w_4)$T′(v2 + v4, w2 + w4).

No, the answer is incorrect.
Score: 0
Accepted Answers:
Suppose Ayesha contributes for herslef and also on behalf of Sabya. Then the contribution per day by Ayesha is given by $T(x_1 + x_3, y_1 + y_3)$T(x1 + x3, y1 + y3) and $T^{'}(v_1 + v_3, w_1 + w_3)$T′(v1 + v3, w1 + w3), for accommodation and food, respectively.
The total contribution (for the whole trip) for the accommodation by Pritha is given by $2T(x_2, y_2)$

2T(x2, y2), which is equal to $T(2x_2, 2y_2)$T(2x2, 2y2).
Suppose Pritha contributes for herself and also on behalf of Wang for food. Then the contribution per day by Pritha for food is given by $T^{'}(v_2, w_2) + T^{'}(v_4, w_4)$T′(v2, w2) + T′(v4, w4), which is equal to $T^{'}(v_2 + v_4, w_2 + w_4)$T′(v2 + v4, w2 + w4).

1 point
The total contribution (accommodation and food) per day can be denoted by:

- [ ] $f(x, y, v, w) = T(x, y) + T^{'}(v, w)$f(x, y, v, w) = T(x, y) + T′(v, w) which is a linear mapping.
- [ ] $f(-4x, -4y, -4v, -4w) = (-4)^4 f(x, y, v, w)$f(−4x, −4y, −4v, −4w) = (−4)4f(x, y, v, w).
- [ ] $f(x, y, v, w) = T(x, y) + T^{'}(v, w)$f(x, y, v, w) = T(x, y) + T′(v, w) which is not a linear mapping.
- [ ] $f(-4x, -4y, -4v, -4w) = -4f(x, y, v, w)$f(−4x, −4y, −4v, −4w) = −4f(x, y, v, w).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y, v, w) = T(x, y) + T^{'}(v, w)$f(x, y, v, w) = T(x, y) + T′(v, w) which is a linear mapping.
$f(-4x, -4y, -4v, -4w) = -4f(x, y, v, w)$f(−4x, −4y, −4v, −4w) = −4f(x, y, v, w).

Check Answers
Your score is: 0/10